



## Genome Note

# Draft genome sequences of clinical mastitis-associated *Enterococcus faecalis* and *Enterococcus faecium* carrying multiple antimicrobial resistance genes isolated from dairy cows



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## ABSTRACT

**Objectives:** The emergence of antimicrobial-resistant and mastitis-associated *Enterococcus faecalis* and *Enterococcus faecium* is of great concern due to the huge economic losses associated with enterococcal infections. Here we report the draft genome sequences of *E. faecalis* and *E. faecium* strains that were isolated from raw milk samples obtained from mastitis-infected cows in Bangladesh.

**Methods:** The two strains were isolated, identified, and genomic DNA was sequenced using the Illumina NextSeq 550 platform. The assembled contigs were analysed for virulence, antimicrobial resistance genes, and multilocus sequence type. The genomes were compared to previously reported *E. faecalis* and *E. faecium* genomes to generate core genome phylogenetic trees.

**Results:** *E. faecalis* strain BR-MHR218Efa and *E. faecium* strain BR-MHR268Efe belonged to multilocus sequence types ST-190 and ST-22, respectively, both of which appear to represent relatively rare sequence types. BR-MHR268Efe harboured only one antibiotic resistance gene encoding resistance towards macrolides (*Isa(A)*), while BR-MHR218Efa harboured ten different antibiotic resistance genes encoding resistance to aminoglycosides (*ant(6)-Ia*, *aph(3')-III*), sulphonamides (*aac(6')-II*), lincosamides (*lnu(B)*), macrolides (*erm(B)*), MLSB antibiotics (*msr(C)*), tetracyclines (*tet(M)*, *tet(L)*), trimethoprim (*dfrG*), and pleuromutilin-lincosamide-streptogramin A (*Isa(E)*). Virulence gene composition was different between the two isolates. BR-MHR218Efa harboured only two virulence genes involved in adherence (*acm* and *scm*). BR-MHR268Efe harboured eight complete virulence operons including three operons involved in adherence (*Ace*, *Ebp pili*, and *EfaA*), two operons involved in biofilm formation (*BopD* and *Fsr*), and three exoenzymes (gelatinase, hyaluronidase, *SprE*).

**Conclusions:** The genome sequences of the strains BR-MHR268Efe and BR-MHR218Efa will serve as a reference point for molecular epidemiological studies of mastitis-associated *E. faecalis* and *E. faecium*. Additionally, the findings will help understand the complex antimicrobial-resistance in livestock-associated *Enterococci*.

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## 1. Introduction

Antimicrobial resistance is a global quintessential *One Health* dilemma that is interconnected within and across human, animal, and environmental health. Gram-negative and Gram-positive bacteria of multiple genera are important causative agents of bovine mastitis throughout the world. Mastitis is a complex, multiaetio-

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logical, and serious dairy cow disease involving udder and intramammary gland inflammation caused by infections from mastitis-causing pathogens [1]. Mastitis poses a risk to public health, veterinary health, and the farming industry worldwide. Mastitis causes devastating economic losses due to reduced milk production and quality, consequent milk withdrawal due to antibiotic use, treatment costs, and culling of diseased cows on dairy farms [2]. Enterococci are ubiquitous environmental bacteria capable of causing opportunistic infections in humans and many animals, including mastitis in dairy cows [3,4]. The detection of mastitis-causing and antimicrobial-resistant *Enterococcus faecalis* and *Enterococcus faecium* isolated from livestock and dairy farm environments has been reported worldwide [5–8]; however, there are very scarce data available on using whole-genome sequencing for the characterization of these mastitis-causing pathogens, including Enterococci.

Extensive use of antibiotics in the management of mastitis has resulted in the emergence and spread of antimicrobial and multidrug-resistant *E. faecalis* and *E. faecium* [9,10].

The aetiologies, management, treatment, and physiological and molecular characterization of mastitis-causing pathogens, as well as the metagenomics of milk microbiota, were recently reported in Bangladesh [6,11]. Recently, we reported on the detection and genome sequences of two *Escherichia coli* strains: one strain was assigned to a novel sequence type 13054 and the other strain belonged to sequence type 101. Both strains were isolated from raw milk samples obtained from lactating cows infected with mastitis in Bangladesh [12].

Here we report on the draft genome sequences of *E. faecalis* and *E. faecium* strains isolated from raw milk samples obtained from July to August 2021 from mastitis-affected lactating cows in Bangladesh.

2. Methods

In the present study, milk samples were collected from 36- and 46-month-old female lactating Holstein Friesian (*Bos taurus taurus*) cows, as previously reported [13]. Briefly, cow udders were washed with clean water and dried, then the udder teats were rubbed with 70% ethanol. The first two strings were discarded, and a California mastitis test was performed to determine somatic cell counts in milk samples. Milk samples (10 mL) were collected in sterile tubes and samples were transported to the laboratory, maintaining a cold chain for further analysis. Somatic cell counts were performed using Lactoscan Combo's SCC (Milkotronic Ltd, Bulgaria) according to the manufacturer's protocol. Nutrient broth (10 mL) was inoculated with 500 µL of the milk sample, incubated at 37°C for 18 h, and subsequently streaked on *Enterococcus* agar plates (HiMedia). The inoculated plates were incubated at 37°C for 24 h and subcultured to isolate pure colonies. Pure single colonies were identified using matrix-assisted laser desorption ionisation time of flight (MALDI-TOF) mass spectrometry (MS) using a MALDI Biotyper (Bruker Daltonics, Bremen, Germany), and isolates with a log score ≥2 indicated species identification, as previously reported [14]. *Enterococcus* isolates were subjected to whole-genome sequencing as previously described [12].

DNA was extracted from overnight cultures grown in nutrient broth at 37°C using a genomic DNA Purification Kit (Promega, USA). Sequencing libraries were prepared using the Nextera XT library preparation kit (Illumina, USA) and sequenced on an Illumina NextSeq 550 using the NextSeq 500/550 High Output Kit v2.5 (300 cycles). Default parameters were used for all software, unless otherwise specified. Quality control was performed using the ASA<sup>3</sup>P pipeline (v1.4.0) [15]. Assembly was performed using SPAdes v3.13.0 [16] integrated in ASA<sup>3</sup>P. Contigs, excluding those smaller than 200 bp, were uploaded to the National Center for

**Table 1**  
Basic characteristics of the whole-genome sequencing, assemblies, annotation, and genes of *Enterococcus faecalis* strain BR-MHR218Efa and *Enterococcus faecium* strain BR-MHR268Efe

| Parameter                   | BR-MHR218Efa | BR-MHR268Efe |
|-----------------------------|--------------|--------------|
| No. of raw reads            | 9 438 233    | 9 307 778    |
| Average read length (nt)    | 149          | 149          |
| Average coverage (x)        | 909          | 816          |
| No. of contigs >200 bp      | 118          | 17           |
| N <sub>50</sub> [bp]        | 70.213       | 305.353      |
| Genome size (bp)            | 2 576 280    | 2 810 324    |
| G+C content (%)             | 38           | 37           |
| No. of genes (total)        | 2534         | 2653         |
| No. of CDSs (with protein)  | 2372         | 2578         |
| No. of genes (RNA)          | 61           | 54           |
| No. of rRNAs (5S, 16S, 23S) | 1, 1, 1      | 1, 1, 1      |
| Complete                    | 1, 1, 1      | 1, 1, 1      |
| Partial                     | 0, 0, 0      | 0, 0, 0      |
| No. of tRNAs                | 54           | 47           |
| No. of ncRNAs               | 4            | 4            |
| No. of pseudogenes (total)  | 101          | 21           |

CDS: coding sequence, ncRNA: non-coding RNA, tRNA: transfer RNA.

Biotechnology Information (NCBI) and annotated using the NCBI Prokaryotic Genome Annotation Pipeline v6.0 [17]. The sequencing, assembly, and annotation data of BR-MHR268Efe and MHR218Efa are summarized in Table 1. Multilocus sequence types (STs) were determined using PubMLST [18], the Homan scheme for *E. faecium* [19], and the Ruiz-Garbajosa scheme for *E. faecalis* [20]. Antimicrobial resistance genes were detected using Resfinder (v4.0) [21]. Virulence genes were detected using the VFDB database [22] included in ASA<sup>3</sup>P. Core genome-based phylogeny was determined using Harvest Suite software tools [23]. The number of single nucleotide polymorphisms (SNPs) was determined using MEGA X [24] and the consensus sequences generated by Harvest Suite. Annotation of the core genome-based phylogenies was generated using ITOL v6 [25] and Inkscape 0.91 (<https://inkscape.org/release/inkscape-0.91/?latest=1>).

3. Results

In the current study, a total of 423 randomly selected lactating cows were tested for the detection of *Enterococci* in milk samples. It was found that 44.68% (189/423) of the cows were mastitis positive, of which 17.49% (74/423) had clinical and 27.19% (115/423) had subclinical mastitis, as previously reported [13]. *Enterococcus* isolation was performed by culture methods using *Enterococcus* agar plates and presumptive isolates were subsequently confirmed using MALDI-TOF MS [14]. It was found that the prevalence rates of *E. faecalis* and *E. faecium* were 14.28% (27 out of 189 samples) and 6.35% (12 out of 189 samples), respectively, in the in the examined milk samples.

**Table 2**  
Antibiotic resistance genes detected in *Enterococcus faecalis* strain BR-MHR218Efa and *Enterococcus faecium* strain BR-MHR268Efe in the present study

| Antibiotic class                           | BR-MHR218Efa                  | BR-MHR268Efe       |
|--------------------------------------------|-------------------------------|--------------------|
| Aminoglycoside                             | <i>ant[6]-Ia, aph(3')-III</i> |                    |
| Sulphonamide                               | <i>aac(6')-II-like</i>        |                    |
| Lincosamide                                | <i>lnu(B)</i>                 |                    |
| Macrolide                                  | <i>erm(B)</i>                 | <i>lsa(A)-like</i> |
| Macrolide, lincosamide and streptogramin B | <i>msr(C)-like</i>            |                    |
| Tetracycline                               | <i>tet(M)-like, tet(L)</i>    |                    |
| Trimethoprim                               | <i>dfrG</i>                   |                    |
| Pleuromutilin-lincosamide-streptogramin A  | <i>lsa(E)</i>                 |                    |

**Table 3**  
Virulence genes detected in *Enterococcus faecalis* strain BR-MHR218Efa and *Enterococcus faecium* strain BR-MHR268Efe in the present study

| Virulence category | Virulence determinant  | Genes present in isolates |                               | Operon       |
|--------------------|------------------------|---------------------------|-------------------------------|--------------|
|                    |                        | BR-MHR218Efa              | BR-MHR268Efe                  |              |
| Adherence          | Acm (VF0419)           | <i>acm</i>                |                               | Complete     |
|                    | Scm (VF0418)           | <i>scm</i>                |                               | Complete     |
|                    | Ace (VF0355)           |                           | <i>ace</i>                    | Complete     |
|                    | AS (VF0352)            |                           | <i>asa1</i>                   | Not complete |
|                    | Ebp pili (VF0538)      |                           | <i>srtC, ebpC, ebpA, ebpB</i> | Complete     |
|                    | EfaA (VF0354)          |                           | <i>efaA</i>                   | Complete     |
| Biofilm            | BopD (VF0362)          |                           | <i>bopD</i>                   | Complete     |
|                    | Fsr (VF0360)           |                           | <i>fsrA, fsrB, fsrC</i>       | Complete     |
| Exoenzyme          | Gelatinase (VF0357)    |                           | <i>gelE</i>                   | Complete     |
|                    | Hyaluronidase (VF0359) |                           | <i>EF0818, EF3023</i>         | Complete     |
|                    | SprE (VF0358)          |                           | <i>sprE</i>                   | Complete     |
| Immune modulation  | Capsule (VF0361)       |                           | <i>cpsA, cpsB</i>             | Not complete |

BR-MHR218Efa and BR-MHR268Efe belonged to multilocus sequence types ST-190 and ST-22, respectively. Both sequence types appear to represent relatively rare sequence types, as only three ST-190 *E. faecium* and six ST-22 *E. faecalis* STs have been deposited in the PubMLST database.

*E. faecium* ST-190 has been isolated from meat chickens in Australia [26] as well as human infections in Portugal [27]. To the authors' knowledge, it has never before been detected in mastitis samples.

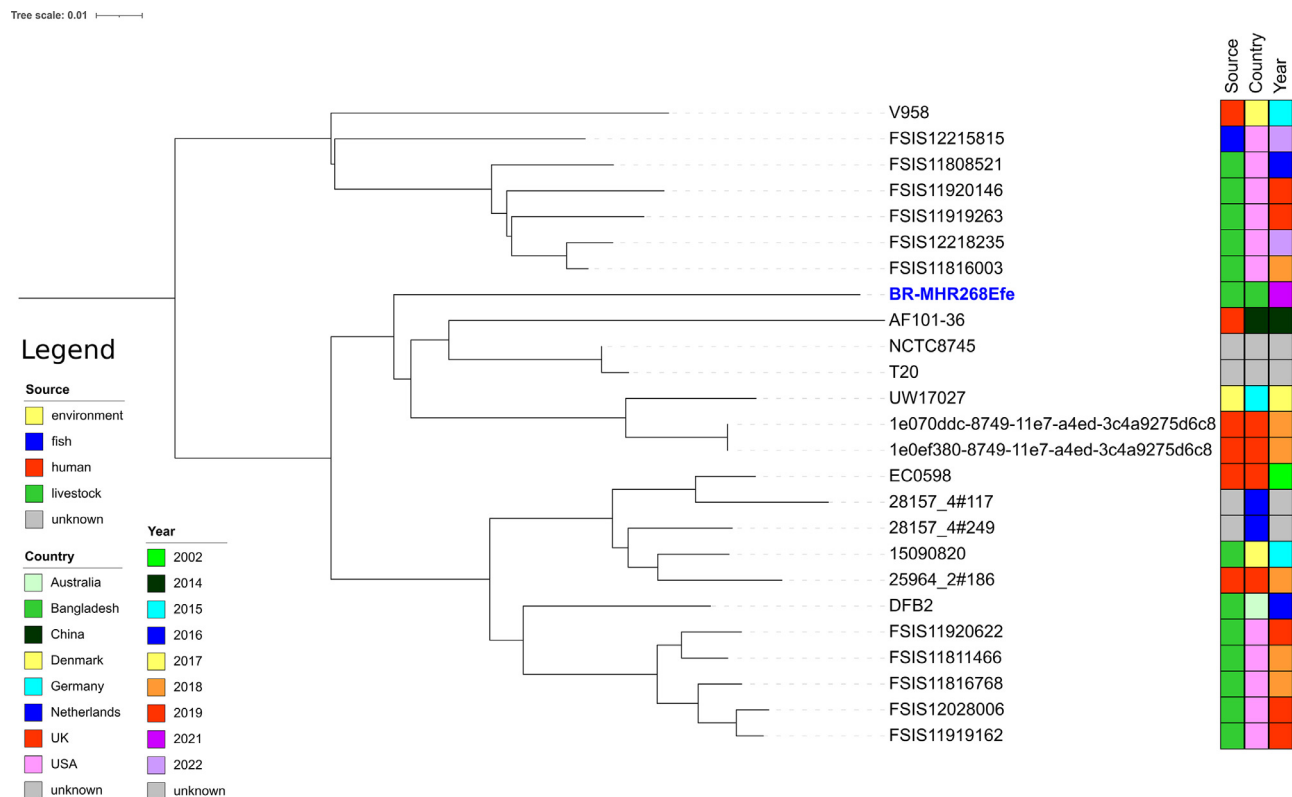
*E. faecalis* ST-22 has been isolated from mastitis samples [28], but also from other samples such as chicken meat and from human infections in Portugal [29]. To the authors' knowledge, *E. faecalis* ST-22 has not been detected in mastitis samples before the current study.

The antibiotic resistance genes detected in the two isolates differed greatly (Table 2). BR-MHR268Efe harboured only one an-

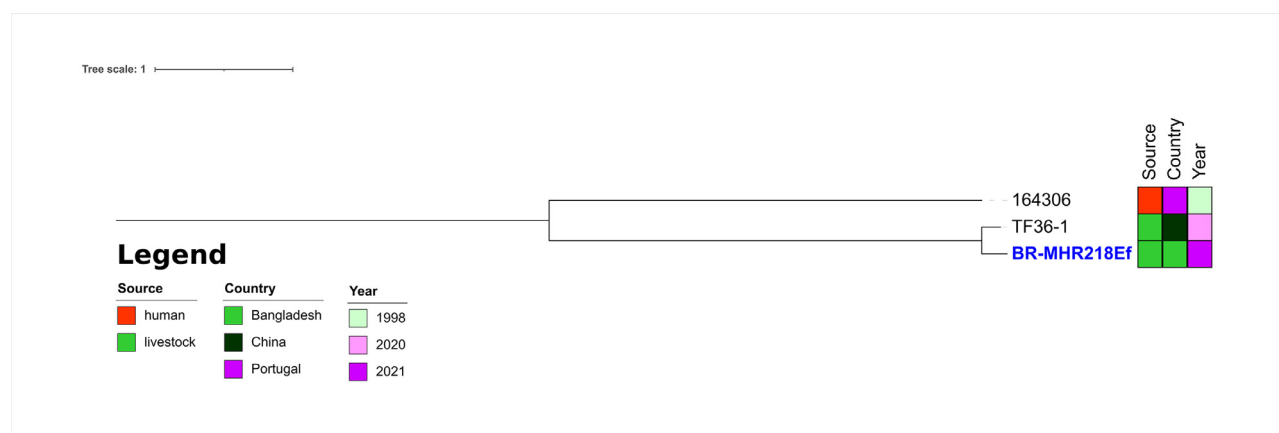
tibiotic resistance gene encoding resistance towards macrolides (*lsa(A)*), while BR-MHR218Efa harboured ten different antibiotic resistance genes encoding resistance to aminoglycosides (*ant(6)-Ia*, *aph(3')-III*), sulphonamides (*aac(6')-II*), lincosamides (*lnu(B)*), macrolides (*erm(B)*), MLSB antibiotics (*msr(C)*), tetracyclines (*tet(M)*, *tet(L)*), trimethoprim (*dfrG*) and pleuromutilin-lincosamide-streptogramin A (*lsa(E)*).

Virulence genes also differed greatly between the two isolates (Table 3). BR-MHR218Efa harboured only two virulence genes involved in adherence (*acm*, *scm*). BR-MHR268Efe harboured eight complete virulence operons (Table 3): three operons involved in adherence (Ace, Ebp pili, and EfaA), two operons involved in biofilm formation (BopD and Fsr), and three exoenzymes (gelatinase, gyaluronidase, and SprE).

As only a limited number of *E. faecium* ST-190 and *E. faecalis* ST-22 were present in the PubMLST database ( $n = 1$  each), we used



**Fig. 1.** Core genome-based analysis of *Enterococcus faecium* strain BR-MHR268Efe and all *Enterococcus faecium* ST-22 isolates publicly available in the National Center for Biotechnology Information assembly database.



**Fig. 2.** Core-genome-based analysis of *Enterococcus faecalis* strain BR-MHR218Efa and all *Enterococcus faecalis* ST-190 isolates publicly available in the National Center for Biotechnology Information assembly database.

a different strategy to determine the closest relatives of the two isolates. All assemblies for *E. faecalis* ( $n = 7276$ ) and *E. faecium* ( $n = 19868$ ) were downloaded from NCBI (as of 13 July 2023) and analysed using ASA<sup>3</sup>P to determine the multilocus sequence type. *E. faecalis* ST-22 ( $n = 24$ , supplementary Table A) and *E. faecium* ST-190 ( $n = 2$ , supplementary Table B) were detected and compared with BR-MHR218Efa and BR-MHR268Efe using core genome-based phylogenetic analysis.

The closest relative to BR-MHR268Efe was NCTC8745 (BioSample SAMEA1046663) from an unknown source (Fig. 1, Supplementary Table C), albeit the relatively high number of SNPs ( $n = 237$ ) indicates a very distant genetic relationship. The closest relative to BR-MHR218Efa was TF36-1 (BioSample SAMN31232721) isolated from an unidentified source from a pig in China (Fig. 2, Supplementary Table D). The very high number of SNPs ( $n = 559$ ) indicates that these isolates are not related.

## 4. Conclusions

Bovine mastitis is a significant zoonotic disease of serious global impact [30]. Mastitis has a zoonotic potential associated with shedding of bacteria and their toxins in unpasteurized milk, which may lead to potential health threats in humans and other animals [5,30]. In addition, antimicrobial resistance from bovine mastitis pathogens may lead to severe infections. The *E. faecium* and *Enterobacter* genus are members of the ESKAPE pathogens, which are among the main bacterial causative agents of antibiotic-resistant nosocomial infections [31,32] that have been designated “priority status” by the World Health Organization [33].

To the authors' knowledge, this is the first report on whole-genome sequencing of mastitis-associated *Enterococci* species in Bangladesh. The present study represents important findings on the detection of mastitis-associated *Enterococci* species in dairy cows, which requires further extensive investigations. This report highlights the significance of continued genomic surveillance of mastitis-associated bacteria, which continue to pose a serious global threat to animal health and food security. Monitoring and genomic characterization of mastitis-associated *Enterococci* species in food-chain cattle and food-production environments will help to further understanding of its zoonotic potential, role in the spread of antimicrobial resistance, and pathogenesis. Furthermore, comprehensive genome analysis of mastitis-associated *Enterococci* species will help us to understand the genomic architecture, rearrangements, and movement of mobile genetic elements, virulence-associated genes, and antimicrobial resistance determinants. It will

also help to elucidate their complex aetiology and ability to cause mastitis in udder-healthy cows.

## Data availability

All isolates reported in this study were submitted to Public Databases for molecular typing and microbial genome diversity for curation and are publicly available in *Enterococcus faecalis* PubMLST and *Enterococcus faecium* PubMLST, under ID numbers 2346 and 4570, respectively. This whole-genome sequencing project has been deposited at DDBJ/ENA/GenBank under the BioProject number PRJNA716986, BioSample accession numbers SAMN26025964 and SAMN26025968, and GenBank accession numbers JALBGN000000000 and JALBGJ000000000 for BR-MHR218Efa and BR-MHR268Efe, respectively. The versions described in this paper are the first version. The sequences have been submitted to the Sequence Read Archive (SRA) under the accession numbers SRR18182113 and SRR18182109.

## Ethics Statement

The study protocol entitled “Development of polyvalent mastitis vaccine and probiotics for prevention of mastitis in cows” under the project entitled “Polyvalent Vaccine Development for Mastitis in Dairy Cow” reference number AWEEC/BAU/2020(44) was approved by the Animal Welfare and Experimentation Ethics Committee, Bangladesh Agricultural University, Mymensingh-2202, Bangladesh.

## Competing interests

None declared.

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## Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.jgar.2024.05.011](https://doi.org/10.1016/j.jgar.2024.05.011).

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